

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:		I					
Paper Category:		DSC4-1 (Major)					
Paper Name:		Practical based on DSC4-1 (Group - II) Paper Code : G06-0104-P					
Credit:		02		Practical:		4 Hrs./Week	
Marks:	UA:	30	CA:	20	Total:	50	

Choose one of the following languages to solve the assignment: Python / C++

1. Write a Python program to prepare equally spaced data for the given arguments.
2. Write a Python program to check whether the given data is equally spaced or not.
3. Write a Python program to prepare Newton's forward difference table. (Both static and Dynamic programmers)
4. Write a Python program to prepare Newton's backward difference table. (Both static and Dynamic programmers)
5. Write a Python program that implements Newton's forward difference interpolation formula to interpolate a value at a given point using equally spaced data points.
6. Write a Python program that implements Newton's backward difference interpolation formula to interpolate a value at a given point using equally spaced data points.
7. Write a Python programme to prepare divided difference table for a set of given data points.
8. Write a Python programme that calculates the absolute error, relative error and percentage error between a true value and an approximate value.
9. Write a Python program that performs addition, subtraction, multiplication, and division of two floating-point numbers in normalized form.
10. Write a Python program to find addition and subtraction of two matrices.
11. Write a Python program to find multiplication of two matrices.
12. Write a Python program to solve system of linear equations by using Jacobi's iterative method.
13. Write a Python program to solve system of linear equations by using Gauss Seidel iterative method.
14. Write a Python program to find the interval in which root of the equation lies.
15. Write a Python program to find one of the roots of given non linear equation by using Bisection method.

16. Write a Python program to find one of the roots of given non linear equation by using Regula Falsi or False Position method.
17. Write a Python program to find one of the roots of given non linear equation by using Newton Raphson Method..